HW0 LA Review in R

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1.

```r
A <- matrix(data = c(8, 6, 6, 8), nrow = 2, byrow = TRUE)
B <- matrix(data = c(1, 8, 3, 5), nrow = 2, byrow = TRUE)
```

(a)

```r
A+B
```

```
     [,1] [,2]
[1,]    9   14
[2,]    9   13
```

(b)

```r
A-B
```

```
     [,1] [,2]
[1,]    7   -2
[2,]    3    3
```

(c)

```r
A%*%B
```

```
     [,1] [,2]
[1,]   26   94
[2,]   30   88
```

(d)

```r
B%*%A
```

```
     [,1] [,2]
[1,]   56   70
[2,]   54   58
```

2.

```r
A <- matrix(data = c(3, 1, 3, 7), ncol = 2, byrow = TRUE)
B <- matrix(data = c(2, 2, 10, 4, 7, 3), ncol = 3, byrow = TRUE)
```

(a)

```r
A+B
```

```
Error in `A + B`:
! non-conformable arrays
```

(b)

```r
A-B
```

```
Error in `A - B`:
! non-conformable arrays
```

(c)

```
A%*%B
```

```
     [,1] [,2] [,3]
[1,]   10   13   33
[2,]   34   55   51
```

(d)

```
B%*%A
```

```
Error in `B %*% A`:
! non-conformable arguments
```

3.

```
A <- matrix(data = c(4, 3, 11, 2), ncol = 2, byrow = TRUE)
B <- matrix(data = c(2, 8, 8, 12), ncol = 2, byrow = TRUE)
C <- matrix(data = c(1, 1, 2, 1, 3, 5, 2, 5, 8), ncol = 3, byrow = TRUE)
D <- matrix(data = c(4, 4, 2, 4), ncol = 2, byrow = TRUE)
```

(a)

```
t(A)
```

```
     [,1] [,2]
[1,]    4   11
[2,]    3    2
```

```
isSymmetric(A)
```

```
[1] FALSE
```

(b)

```r
t(B)
```

```
     [,1] [,2]
[1,]    2    8
[2,]    8   12
```

```r
isSymmetric(B)
```

```
[1] TRUE
```

(c)

```r
t(C)
```

```
     [,1] [,2] [,3]
[1,]    1    1    2
[2,]    1    3    5
[3,]    2    5    8
```

```r
isSymmetric(C)
```

```
[1] TRUE
```

(d)

```r
t(D)
```

```
     [,1] [,2]
[1,]    4    2
[2,]    4    4
```

```r
isSymmetric(D)
```

```
[1] FALSE
```

4.

(a)

```r
solve(A)
```

```
      [,1]  [,2]
[1,] -0.08  0.12
[2,]  0.44 -0.16
```

(b)

```r
solve(B)
```

```
     [,1]  [,2]
[1,] -0.3  0.20
[2,]  0.2 -0.05
```

(c)

```r
solve(A%*%B)
```

```
       [,1]   [,2]
[1,]  0.112 -0.068
[2,] -0.038  0.032
```

(d)

```r
solve(B) %*% solve(A)
```

```
       [,1]   [,2]
[1,]  0.112 -0.068
[2,] -0.038  0.032
```

```r
(solve(B) %*% solve(A)) == (solve(A%*%B))
```

```
      [,1] [,2]
[1,] FALSE TRUE
[2,]  TRUE TRUE
```

Funnily enough, R thinks the results are different (my guess is some very minor numeric computation error in cell [1,1]), but yes, just looking at the two results you can see they are equal.

5.

```r
A <- matrix(data = c(4, 3, 11, 2), ncol = 2, byrow = TRUE)
B <- matrix(data = c(2, 8, 4, 0, 5, 333, 1, 0, 1, 1, 7, 0, 6, 10, 423, 0), ncol = 4, byrow =
C <- matrix(data = c(1, 1, 2, 1, 3, 5, 2, 5, 8), ncol = 3, byrow = TRUE)
D <- matrix(data = c(4, 0, 0, 2), ncol = 2, byrow = TRUE)
```

(a)

```r
det(A)
```

```
[1] -25
```

(b)

```r
det(B)
```

```
[1] 0
```

(c)

```r
det(C)
```

```
[1] -1
```

```r
c_transpose <- t(C)
det(c_transpose)
```

```
[1] -1
```

Verified that $|C^\top| = |C|$

(d)

```r
det(D)
```

```
[1] 8
```

```r
d_inv <- solve(D)
det(d_inv)
```

```
[1] 0.125
```

Verified that $|D^{-1}| = 1/|D|$

6.

```r
A <- matrix(data = c(7, 2, 5, 8, -3, 4, 0, 2, 9, 3, 6, 5, 3, 1, 2, 1), ncol = 4, byrow = TRUE

A11 <- A[1:2, 1:3]
A12 <- A[1:2, 4]
A21 <- A[3:4, 1:3]
A22 <- A[3:4, 4]

A11
```

```
     [,1] [,2] [,3]
[1,]    7    2    5
[2,]   -3    4    0
```

```
A12
```

```
[1] 8 2
```

```
A21
```

```
     [,1] [,2] [,3]
[1,]    9    3    6
[2,]    3    1    2
```

```
A22
```

```
[1] 5 1
```